# Spring potential energy

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Calculate stored spring energy and distinguish it from force.

**SUCCESS CRITERIA - I can...**

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A pendulum loses 15 J gravitational energy while descending. Neglect losses. Find its kinetic-energy gain.

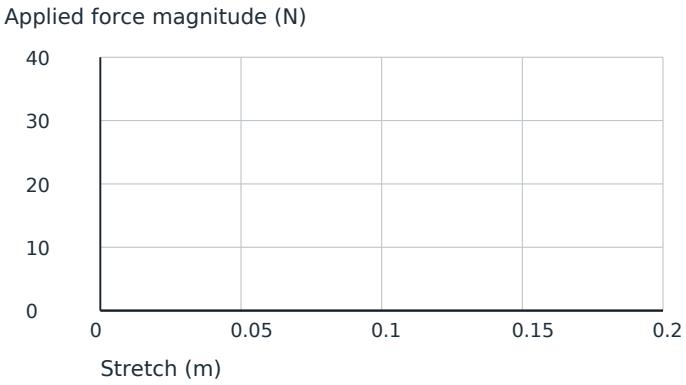
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

**Worked reasoning:** A spring with k=200 N/m stretched 0.10 m stores $\frac{1}{2}kx^2$=1 J. Doubling stretch to 0.20 m stores 4 J, not 2 J.

**Your turn:** Draw force-versus-stretch triangles and compare their areas.




**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

The same spring is displaced x and 2 x. Compare stored energies and force magnitudes.



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Use k=100 N/m and x=0.20 m. Find $x^2$, then $U=\frac{1}{2}\times100\times x^2$.



Mark equilibrium and displacement x.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

Why is energy positive for both positive and negative x even though spring force reverses?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A 4 N friction force opposes a 6 m displacement. Find work, including sign and units.
Label force and displacement, including their relative direction.

Add force and displacement arrows; label the angle.



Force: ____ N  Displacement: ____ m

**Marzano check - circle:** 1 2 3

## 07 Exit check

A spring stores 3 J at a certain compression. How much does it store at double compression?

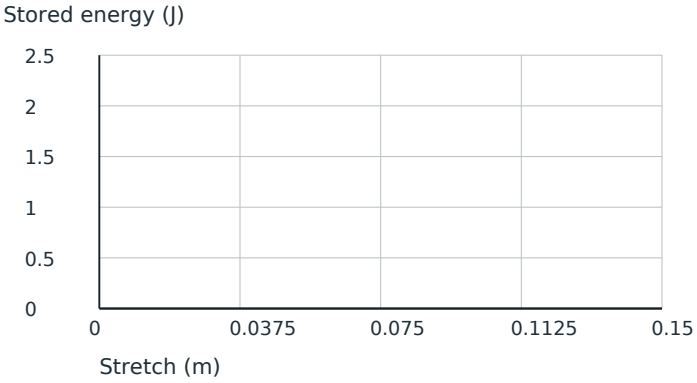


Mark equilibrium and displacement x.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Find the energy for stretches 0.05, 0.10, 0.15 m when k=200 N/m. Sketch the energy trend.




**Marzano check - circle:** 1 2 3